

Cambridge International AS & A Level

MATHEMATICS**9709/12**

Paper 1 Pure Mathematics 1

May/June 2025**MARK SCHEME**

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Due to a series-specific issue during the live exam series, all candidates were awarded full marks for question 6. The mark scheme for these questions was not used by examiners.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **22** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mathematics-Specific Marking Principles

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	More information required
	Accuracy mark awarded zero
	Accuracy mark awarded one
	Independent accuracy mark awarded zero
	Independent accuracy mark awarded one
	Independent accuracy mark awarded two
	Benefit of the doubt
	Blank Page
	Incorrect
Dep	Used to indicate DM0 or DM1

Annotation	Meaning
DM1	Dependent on the previous M1 mark(s)
	Follow through
	Indicate working that is right or wrong
Highlighter	Highlight a key point in the working
	Ignore subsequent work
	Judgement
	Judgement
	Method mark awarded zero
	Method mark awarded one
	Method mark awarded two
	Misread
	Omission or Other solution
Off-page comment	Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to.
On-page comment	Allows comments to be entered in speech bubbles on the candidate response.
	Judgment made by the PE
	Premature approximation
	Special case
	Indicates that work/page has been seen

Annotation	Meaning
	Error in number of significant figures
	Correct
	Transcription error
	Correct answer from incorrect working

PUBLISHED**Mark Scheme Notes**

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

- M** Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).
- B** Mark for a correct result or statement independent of method marks.
- DM or DB** When a part of a question has two or more ‘method’ steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.
- FT** Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.
- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
 - For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
 - The total number of marks available for each question is shown at the bottom of the Marks column.
 - Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
 - Square brackets [] around text or numbers show extra information not needed for the mark to be awarded.

Abbreviations

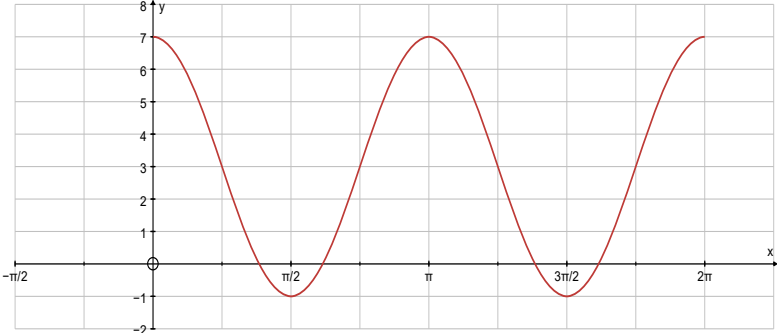
AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no ‘follow through’ from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

Question	Answer	Marks	Guidance
1	{Stretch} {factor 2} {'parallel to y -axis' or 'in y -direction' or 'vertically'}	B2,1,0	B2 for 3 correct components. B1 for 2.
	{Translation} $\begin{pmatrix} \{0\} \\ \{-14\} \end{pmatrix}$ or $\{-14\}$ {'parallel to the y -axis' or 'in the y -direction' or 'vertically'.}	B2,1,0	B2 for 3 correct components. B1 for 2.
	Alternative Method for Question 1		
	{Translation by} $\begin{pmatrix} \{0\} \\ \{-7\} \end{pmatrix}$ or $\{-7\}$ {'parallel to the y -axis' or 'in the y -direction' or 'vertically'}	B2,1,0	B2 for 3 correct components. B1 for 2.
	{Stretch} {factor 2} {'parallel to y -axis' or 'in y -direction' or 'vertically'}	B2,1,0	B2 for 3 correct components. B1 for 2.
		4	

Question	Answer	Marks	Guidance
2	$2\left(\frac{-y-4}{2}\right)y + 5y^2 = 24$ or $2\left(\frac{24-5y^2}{2y}\right) + y + 4 [= 0]$ or $2x(-4-2x) + 5(-4-2x)^2 = 24$	*M1	OE For eliminating x or y . Condone sign errors in the rearrangements.
	$4y^2 - 4y = 24$ or $16x^2 + 72x + 56 [= 0]$	DM1	OE For simplifying to a 3-term quadratic; terms do not all have to be on the same side. Condone sign errors in the expansions or in the collecting of terms.
	$\Rightarrow y = -2, y = 3$ or $x = -1, x = -\frac{7}{2}$	B1	OE Not for a correct (x, y) pair. Allow from a correct quadratic (ignore any working seen).
	$(-1, -2), \left(-\frac{7}{2}, 3\right)$	B1	OE Allow from a correct quadratic (ignore any working seen). Condone $x = -1$ and $y = -2$ and $x = -\frac{7}{2}$ and $y = 3$.
		4	

Question	Answer	Marks	Guidance
3	$\binom{5}{3}$ or $\binom{5}{2}$ or $10 \times (px^2)^{5-3} \times \left(\frac{4}{p}x\right)^3$	B1	May be seen in a list. Allow with or without x 's. Condone missing brackets only if recovered later.
	$10p^2 \times \frac{64}{p^3}$	B1	This term must now be identified if given as part of a list. Allow with or without x 's.
	$\frac{\text{Their integer } k}{p} [x^7] = 1280[x^7]$	M1	Condone e.g. $\frac{640x^7}{p} = 1280$ if x^7 then disappears in subsequent work. Must be from $a \times (px^2)^{5-3} \times \left(\frac{4}{p}x\right)^3$.
	$p = \frac{1}{2}$	A1	OE SC B1 for $\binom{5}{4}$ or $\binom{5}{1}$ or $5 \times (px^2)^{5-1} \times \left(\frac{4}{px}\right)^1 \Rightarrow p = 4$.
		4	

Question	Answer	Marks	Guidance
4(a)	$\left[\frac{dy}{dx}\right] = \frac{3}{2}ax^{\frac{1}{2}} - 12$	*M1	For attempt at differentiation; at least one correct term needed. Condone poor notation throughout.
	$\left[\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt} = \right] \left(\frac{3}{2}a \times 9^{\frac{1}{2}} - 12 \right) \times 5$	DM1	For correct use of chain rule with 5, $x=9$ and <i>their</i> $\frac{dy}{dx}$. Condone missing brackets and allow errors in their working.
	$\left[\frac{dy}{dt} = \right] 5\left(\frac{9}{2}a - 12\right)$ or $\frac{45}{2}a - 60$ or $22.5a - 60$ or $\frac{45a - 120}{2}$ or $\frac{15}{2}(3a - 8)$	A1	OE simplified form.
		3	
4(b)	$\frac{3}{2}a \times \left(\frac{1}{4}\right)^{\frac{1}{2}} - 12 = 0$	M1	For setting <i>their</i> 2 term $\frac{dy}{dx}$ with at least one term correct $= 0$ and substituting $x = 0.25$. Condone missing brackets. Allow a restart for $\frac{dy}{dx}$ if 2 terms seen and at least one term correct.
	$\Rightarrow a = 16$	A1	
		2	

Question	Answer	Marks	Guidance
5(a)	[Greatest] 7, [least] -1	B1B1	B2 for answers of -1 and 7 only. B1 for one correct value. Ignore incorrect identification/inequality.
		2	
5(b)		B2,1,0	Ignore any graph outside domain $(0, 2\pi)$. B1 for two complete cycles, one from 0 to approximately π and the other finishing at approximately 2π. Starting at <i>their</i> greatest value and initially decreasing. Condone straight lines and minimum above the x-axis or joining points with straight lines. B2 for correct curve; condone any incorrect x-axis intercepts. Graph must start to level off at both 0 and 2π. Ignore any y-labels, but the curve should be more above the x-axis than below.
		2	
5(c)	3 [solutions]	B1	Ignore any graphs drawn
		1	

Question	Answer	Marks	Guidance
6	$6 - 3x = 3 \Rightarrow x\text{-coordinate of point of intersection} = 1$	B1	
	$\left[\frac{18}{5}(5x+4)^{\frac{1}{2}} \right]_0^1$	B1 B1	B1 for $k(5x+4)^{\frac{1}{2}}$
	Area = $\frac{18}{5} \times 3 - \frac{18}{5} \times 2 \left[= \frac{18}{5} \right]$	M1	
	Line crosses x -axis at $x = 2$, so triangle area = $\frac{1}{2} \times 3 \times 1$	M1	
	Total area = $\frac{51}{10}$	A1	
		6	

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Question	Answer	Marks	Guidance
7(a)	For appropriate use of $\tan \theta = \frac{\sin \theta}{\cos \theta}$ or $\tan \theta \cos \theta = \sin \theta$ at least once	M1	The first two marks can be applied in the reverse order but as soon as an error occurs, no further marks are awarded. Candidates can work from LHS to RHS or RHS to LHS for full marks. If they work on both sides simultaneously, maximum of M1M1 only if a common correct expression is reached. Condone missing brackets. If the numerator and denominator are worked on separately, they need to be brought together for the second mark.
	For appropriate use of $\sin^2 \theta + \cos^2 \theta = 1$ or $1 + \tan^2 \theta = \sec^2 \theta$	M1	
	Fully correct proof	A1	WWW Condone missing brackets if recovered.
		3	
7(b)	$\frac{\tan \theta + 7}{\tan^2 \theta - 3} = \frac{5}{\tan \theta} \Rightarrow \tan \theta (\tan \theta + 7) = 5(\tan^2 \theta - 3)$	*B1	OE Using part (a) and eliminating fractions.
	$\Rightarrow 4 \tan^2 \theta - 7 \tan \theta - 15 = 0 \Rightarrow (4 \tan \theta + 5)(\tan \theta - 3) = 0$	DM1	Factorising or other accepted method for solving <i>their</i> 3-term quadratic in $\tan \theta$. Condone errors made in forming 3-term quadratic.
	$\tan \theta = -1.25$ and $\tan \theta = 3$	B1	OE Independent of factorisation of the quadratic. And no extra solutions. Allow e.g. $x = \dots$ provided $x = \tan \theta$ is seen. Can be implied by a correct final answer.
	$\theta = 71.6^\circ$ and $\theta = 128.7^\circ$	B1	AWRT No others in range $0 \leq \theta \leq 180^\circ$. Ignore any answers outside this range. Maximum 3/4 if incorrect factorisation or formula.
		4	

Question	Answer	Marks	Guidance
8(a)	(2, -2)	B1	Condone $x = 2$ and $y = -2$ if seen together.
	(12, -2)	B1	Condone $x = 12$ and $y = -2$ if seen together.
	(7, -4)	B1	
		3	If B0 B0 for the first two marks, then SC B1 available for $x = 2$ and $x = 12$.
8(b)	$\tan\left(\frac{1}{2}\theta\right) = \frac{5}{2} \left[\Rightarrow \theta = 2 \tan^{-1} \frac{5}{2} \right]$ or $\sin\left(\frac{\theta}{2}\right) = \frac{5}{\sqrt{29}}$ or $\cos\left(\frac{\theta}{2}\right) = \frac{2}{\sqrt{29}}$ or $\theta = \pi - 2 \times \tan^{-1} \frac{2}{5}$ or $\pi - 2 \times 0.381$ or $10^2 = 29 + 29 - 2 \times \sqrt{29} \times \sqrt{29} \cos \theta \left[\Rightarrow \cos \theta = \frac{29 + 29 - 100}{2 \times \sqrt{29} \times \sqrt{29}} = \frac{-21}{29} \right]$	M1	Or other correct method for an isosceles triangle using their A and B of the form $(\quad, -2)$ and C of the form $(\quad, -d)$, where $-d < -2$. Note: $\tan \theta = \frac{5}{2} \Rightarrow 0/2$ unless θ is then doubled. Using $r = 29$ scores 0/2.
	$[\theta =] 2.38$	A1	AWRT Final answer of 136.4° scores max 1/2. Ignore degree symbol if present.
	Alternative Method for Question 8(b)		
	$\tan \theta = \frac{\frac{-2}{5} - \frac{2}{5}}{1 + \frac{-2}{5} \times \frac{2}{5}} \left[= -\frac{20}{21} \right]$	M1	Use of $\tan \theta = \frac{m_2 - m_1}{1 + m_2 m_1}$, where m_1 and m_2 are the gradients of <i>their</i> AC and BC , where A , B and C are of the required form.
	$[\theta =] 2.38$	A1	AWRT Ignore degree symbol if present.
		2	

Question	Answer	Marks	Guidance
8(c)	[Length $AC = \text{Length } BC = \sqrt{5^2 + 2^2}] = \sqrt{29} [=5.385\dots]$	B1	Sight of $\sqrt{29}$. SOL. Condone 5.4. Could be found in parts (a) or (b), but do not award unless it is seen in part (c).
	[Area of large sector =] $\frac{1}{2} \times \sqrt{29} \times \sqrt{29} \times (2\pi - \text{their } \theta) [=56.587\dots]$ or [Area of small sector =] $\frac{1}{2} \times \sqrt{29} \times \sqrt{29} \times \text{their } \theta [=34.518\dots]$	M1	Use of correct sector formula with <i>their</i> identified radius (e.g. 29) and <i>their</i> $(2\pi - \theta)$ or <i>their</i> θ . May be embedded as part of the segment formula.
	Area of triangle = $\frac{1}{2} \times \sqrt{29} \times \sqrt{29} \times \sin \theta$ or $\frac{1}{2} \times 10 \times 2 [=10]$	M1	Use of correct triangle formula with <i>their</i> identified radius (e.g. 29) and <i>their</i> θ . May be embedded as part of the segment formula.
	[Segment area =] 66.6	A1	AWRT [Area of circle – smaller segment = $\pi \times 29 - 34.52 + 10$]
		4	

Question	Answer	Marks	Guidance
9(a)	$\left[\frac{dy}{dx} = \right] \frac{-24x^{-2}}{-2} \left[= \frac{12}{x^2} \right] [+c]$	B1	May be unsimplified.
	$\frac{12}{(-2)^2} + c = 0 \Rightarrow c = -3$	M1	Substituting $x = -2$ into <i>their</i> $\frac{dy}{dx}$ of the form $kx^{-2} + c$, and setting = 0. Condone missing bracket.
	$\left[\frac{dy}{dx} = \right] \frac{12}{x^2} - 3$ or $12x^{-2} - 3$	A1	$\frac{-24}{-2}$ must be simplified to 12, but accept $c = -3$ as the final answer.
		3	

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Question	Answer	Marks	Guidance
9(b)	$\left[\frac{12}{x^2} - 3 = 0 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2 \Rightarrow \right] x = 2$	B1	May be done by inspection or calculator, but must be WWW. Condone ± 2 .
	$\left[\frac{d^2y}{dx^2} = \right] -\frac{24}{(2)^3}$ or -3 or $< 0 \Rightarrow$ maximum	B1	WWW. Or other valid method. Ignore any working for $x = -2$.
		2	
9(c)	$[y =] -12x^{-1} + \text{their } cx \left[= -\frac{12}{x} - 3x \right] [+d]$	B1FT	May be unsimplified. FT <i>their</i> nonzero c .
	$19 = -\frac{12}{-2} - 3 \times (-2) + d \quad [\Rightarrow d = 7]$	M1	Substituting $(-2, 19)$ into their integrated expression (defined by having at least one correct power) and including $+d$.
	$y = -\frac{12}{x} - 3x + 7$	A1	Accept $d = 7$ as final answer if $y = \dots$ seen previously. Allow ' $f(x) =$ ' in place of ' $y =$ '.
		3	

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Question	Answer	Marks	Guidance
9(d)	$\frac{12}{x^2} - 3 = -\frac{9}{4} \Rightarrow \frac{12}{x^2} = \frac{3}{4} \Rightarrow x = \dots \left[\Rightarrow x^2 = 16 \Rightarrow x = 4, y = -8 \right]$	*M1	Setting <i>their</i> $\frac{dy}{dx}$ of the form $kx^{-2} + c = -\frac{9}{4}$, and correctly solving for x from their equation where $x^2 > 0$.
	Gradient of normal = $\frac{4}{9}$	B1	
	$\frac{y+8}{(x-4)} = \frac{4}{9}$	DM1	For using <i>their</i> stated positive x and <i>their</i> stated y and a changed gradient to correctly form a line equation; condone $-\frac{4}{9}$ or $\frac{9}{4}$.
	$4x - 9y - 88 = 0$	A1	OE in correct form.
		4	

Question	Answer	Marks	Guidance
10(a)(i)	$8k - k^2 = k^2 - 4k$ or $2k^2 = 4k + 8k$ or $4k + 2(k^2 - 4k) = 8k$	M1	OE Forming an equation in k only, clearly using the first 3 terms of an AP.
	$[2k(k-6) = 0 \Rightarrow] k = 6$ [or $k = 0$]	A1	Condone extra 'solution' $k = 0$. Note: can be done by inspection or with no working. 2 / 2
		2	

Question	Answer	Marks	Guidance
10(a)(ii)	$d [= 8k - k^2 \text{ or } 2k \text{ or } k^2 - 4k] = 12 \text{ and } a [= 4k] = 24$	*M1	Using a correct method to find a and d from <i>their</i> k . SOI. May have been found in (a)(i) but must be used in (ii) .
	$S_{20} = \frac{20}{2}(2 \times 24 + 19 \times 12) \text{ or } = \frac{20}{2}(24 + 252)$	DM1	Use of correct sum formula with $n = 20$ and <i>their</i> a and d .
	$S_{20} = 2760$	A1	
		3	
10(b)	$ar^3 = 36 \text{ and } ar^5 = 6$ or $r^2 = \frac{1}{6}$	B1	SOI WWW
	$r = \sqrt{\frac{6}{36}} \left[= \frac{1}{\sqrt{6}} \text{ or } \frac{\sqrt{6}}{6} \right]$	*M1	Using a correct method to find r . Condone $\pm$.
	$a = \frac{36}{\left(\frac{1}{\sqrt{6}}\right)^3} \left[= 36 \times \sqrt{6}^3, 216\sqrt{6} \text{ or } \frac{1296}{\sqrt{6}} \right]$	*M1	Using a correct method to find a . Condone $\pm$.
	$S_{\infty} = \frac{216\sqrt{6}}{1 - \frac{1}{\sqrt{6}}}$	DM1	Using the correct formula with <i>their</i> a and <i>their</i> $ r < 1$.
	$[S_{\infty} =] \frac{1296}{\sqrt{6} - 1} \text{ or } \frac{7776}{\sqrt{216} - 6}$	A1	OE in the required form. $r = 0.408 \Rightarrow a = 529 \Rightarrow S_{\infty} = 894$ scores 4/5.
		5	

Question	Answer	Marks	Guidance
11(a)	$\{(x+2)^2\}\{-2\}$	B1B1	B1 for each correct $\{ \}$. Allow $a = 2$, $b = -2$. If contradictory, give preference to the expression.
		2	
11(b)(i)	$y = (x+2)^2 - 2 \Rightarrow y+2 = (x+2)^2$	*M1	Equating y or $f^{-1}(x)$ or f^{-1} or $f(x)$ to <i>their</i> completed square form and first step. x and y may be interchanged at this stage. Condone $\pm$ errors.
	$x = [\pm]\sqrt{y+2} - 2$	DM1	Condone $\pm$ errors during simplification.
	$[f^{-1}(x) =] - \sqrt{x+2} - 2$	A1	Do not condone $x =$ or $f(x) =$.
		3	

Question	Answer	Marks	Guidance
11(b)(ii)	$[gf(x) =] - \{their \text{ completed square form} \} - 4$ or $-(x^2 + 4x + 2) - 4$	*M1	Using $fg(x) = \{(-x - 4) + 2\}^2 - 2$ scores 0/4.
	$[gf(x) =] - (x + 2)^2 - 2$	A1	
	$x = [\pm] \sqrt{-y - 2} - 2$	DM1	Finding x from <i>their</i> completed square form, which must contain a $-(x + k)^2$ term. Condone $\pm$ errors only during simplification. $x = -\sqrt{y + 2} - 2$ is DM0 (Square rooting then $\div$ or $\times -1$) x and y may be interchanged at this stage.
	$[(gf)^{-1}(x) =] - \sqrt{-x - 2} - 2$	A1	Do not condone $x =$.
	Alternative Method for Question 11(b)(ii)		
	$(gf)^{-1} = f^{-1}g^{-1}$	*M1	SOI Allow with <i>their</i> f^{-1} and <i>their</i> g^{-1} . Using $g^{-1}f^{-1}(x)$ scores 0/4.
	$g^{-1}(x) = -x - 4$	A1	
	$(gf)^{-1}(x) = -\sqrt{-x - 4 + 2} - 2$	DM1	Allow with <i>their</i> f^{-1} and <i>their</i> g^{-1} .
	$[(gf)^{-1}(x) =] - \sqrt{-x - 2} - 2$	A1	Do not condone $x =$
		4	